Exercise 3E

3) $F = G \times \frac{m_1 \times m_2}{r^2}$

4) It depends on the distance seperating them because the magnitude of the Force of attraction between the two bodies is indirectly proportional to the distance seperating them.

6) Gravitational constant (G) is numerically equal to the magnitude of force of attraction between two masses each of 1kg placed at a seperation of 1m.

7) The value of $G = 6.67 \times 10^{-11}$
S.I unit of $G = Nm^2 Kg^{-2}$

14) g and G are related to each other because to calculate g we need the value of G.

so to calculate g of an object of mass 1kg, we take the following values:

$m_1 = 1$ kg
$m_2 = (5.96 \times 10^{24})$
$r = (6.37 \times 10^6)$
$G = 6.67 \times 10^{-11}$

So

$$g = \frac{(6.67 \times 10^{-11}) \times (5.96 \times 10^{24}) \times 1}{(6.37 \times 10^6)^2}$$

$= 9.8$ N

So therefore, g and G are related.

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15) As we know the formula for distance =

$$S = \frac{1}{2}gt^2$$

we can use that here by denoting the distance with the letter 'h' for height.

So

$$h = \frac{1}{2}gt^2$$

17) (a) The mass of an object is the ~~number~~ quantity of matter it contains.

(b) The weight of a body is the force with which the earth attracts it. In other words, the weight of a body is the force of gravity on it.

18)

	Mass	Weight
⇒	It is a measure of the quantity of matter contained in the body at rest.	It is the force with which the earth attracts a body
⇒	It is a scalar quantity	It is a vector quantity
⇒	Its S.I unit = kg	Its S.I unit = N
⇒	It is measured by a physical or beam balance	It is measured by a spring balance which is calibrated to read in N.
⇒	It is constant for a body and does not change with the change in place	It is not constant for a body, but varies from place to place due to change in value of g

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